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Project Title:	Characteristics of Light Exposure Necessary for Development of Optimal Countermeasures to Facilitate Circadian Adaptation and Enhance Alertness and Cognitive Performance in Space		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
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Comments:			
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
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Task Description:	Space missions often expose crew members to unusual light-dark cycles that lead to misalignment between the circadian pacemaker and the sleep-wake schedule, resulting in disturbed sleep and impaired waking function. An effective countermeasure is required to facilitate more rapid adaptation of the circadian system and directly enhance alertness and performance. Recently, we and others have shown that short-wavelength (blue) light exposure at night is the most effective wavelength for phase-shifting the circadian pacemaker, suppressing pineal melatonin, enhancing subjective alertness and performance, and increasing brain activation. These data are consistent with stimulation of a novel blue-light sensitive non-rod, non-cone melanopsin-based photoreception system. These effects, however, have only been examined during the night using relatively bright light and recent evidence suggests that the spectral sensitivity of the circadian photoreception system may depend on the time of day, intensity of light, and duration of light. Based on model fits to irradiance-response curves to 460 and 555 nm, we predict that photopic (cone) photoreceptor system, which is maximally sensitive to green (555 nm) light, can contribute significantly to these responses and may be more effective at shifting the circadian system and suppressing melatonin than blue light at low light intensities. Furthermore, while blue light therapy to facilitate circadian adaptation and alertness has been proposed based largely on the effects of melatonin suppression at night, these data may not be transferable to the effects at other times of day (when melatonin is not produced) or for other effects of light. Before blue light is operationalized, three fundamental questions remain: 1) Does the spectral sensitivity depend on light intensity? 2) Do all non-visual effects of light have the same spectral sensitivity? 3) Is the spectral sensitivity of the circadian photoreception system different between the day and the night? Failure to address these questions may result in deployment of suboptimal lighting countermeasures. The proposed project will test whether the spectral sensitivity of the circadian photoreception system is intensity-dependent, response-dependent, or time-of-day dependent. We aim to conduct three experiments: Experiment 1 aims to compare the melatonin suppression, phase resetting and alerting effects of exposure to dim 460 nm and 555 nm monochromatic light exposure at night, and compare these data to historic bright light studies using the same protocol. Experiment 2 will test the effects of bright 460 nm and 555 nm monochromatic light on daytime alertness and compare these data to previously collected night-time data using the same protocol. Experiment 3 has changed from that originally proposed based on state-of-the-art unpublished findings on the effects of light pattern, duration, and wavelength. The revised experiment will test the hypothesis that an intermittent exposure to monochromatic 555 nm light for 6.5 hours during the biological night would cause a greater circadian phase shifting response than exposure to an equal photon density of continuous monochromatic 555 nm light. The data have been combined with those from historical experiments and the results of Experiment 1 to examine the interaction between light timing (day vs night), intensity (dim vs bright), and wavelength (460 vs 555 nm). In brief, as hypothesized, 555 nm dim night-time light exposure caused a significantly greater circadian phase shift as compared to 460-nm light exposure. In addition, there was significantly greater melatonin suppression induced by dim 555-nm light exposure as compared to 460-nm light exposure during the first quarter of the 6.5-h light exposure, but not over the total 6.5-h duration. The acute effects of light on alertness and performance were not consistent with the phase resetting responses. Dim 460-nm night-time light-exposure was associated with significantly quicker reaction times and lapses of attention as compared to dim night-time 555-nm exposure suggesting short-wavelength sensitivity for the acute alerting effects of light. Subjective sleepiness levels showed a similar trend but the results did not reach significance, suggesting that subjective ratings of sleepiness may not be an appropriate way to rate the potential performance risk associated with sleepiness and sleep deprivation. Comparisons between daytime and nighttime bright monochromatic light revealed novel results. Nighttime exposure to bright 460-nm light maintains alertness at levels that approach daytime alertness, suggesting that blue light is a powerful countermeasure to the strong circadian drive for sleepiness at night. Conversely, neither bright nor dim 555 nm light at night could improve performance to the same extent as blue light and has minimal effect on the circadian rhythm of sleepiness. Green 555 nm light may still be effective under some circumstances, however. When bright green light exposure is used during the daytime, 555 nm light appears to improve alertness as the exposure-duration increases (>3-h) although never to the extent that bright blue light maintains alertness, which remains high throughout the entire duration. Green 555 nm light may also be able to exert a temporary alerting effect at the start of a 10-minute vigilance task but cannot maintain a prolonged response for the entire duration, unlike 460 nm exposures. As a result of this work, we envisage the development of 'smart' lighting systems that will detect the ambient lighting conditions using an in-built sensor and automatically adjust the intensity, wavelength and pattern of light emitted to maximize the circadian, alerting, and visual effects of light under any conditions and at any time of day, and at the greatest energy efficiency. Steps to develop such lighting recommendations for the International Space Station have been initiated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The purpose of the proposed studies was to test three specific hypotheses aimed at evaluating the spectral and temporal sensitivity of light to reset the circadian pacemaker and acutely improve alertness and performance during both day-time and night-time exposures. These studies are designed to provide fundamental information about the photoreceptor systems that detect light for a range of circadian, neuroendocrine, and neurobehavioral responses to light in order that optimal lighting can be designed to alleviate circadian misalignment and acutely improve alertness during space exploration missions. A major impact of this research has been to support the development of Revised Lighting Requirements (RLR) for lighting aboard the International Space Station in order to replace the current fluorescent General Luminaire Assemblies (GLA) which will be exhausted in 2015. Results from our current project have provided scientific evidence on the contribution of the novel non-rod, non-cone photoreceptor mechanism used to detect light for non-visual responses including circadian phase resetting and improving alertness. Based on these and other results, the RLR outline how light wavelength and intensity can be adjusted quickly by crew members to optimize lighting for both crew vision and to provide a crew health countermeasure for circadian entrainment, sleep shifting, enhancing alertness and performance (see Flight Transition Plan for details). The Solid State Lighting Assemblies (SSLAs) proposed to replace the GLAs were developed at Kennedy Space Center and represent the first 'smart' lighting system to provide light of different intensities and wavelength specifications to optimize the lighting to operational needs from a single light source. Based on the RLR for use of the SSLAs, we have recently developed a Dynamic Lighting Schedule (DLS) which determines when each of the three pre-determined light settings will be used to optimize lighting for different operational needs throughout the day and to facilitate circadian adaptation, improve alertness and performance, and enhance sleep while maintaining high visual acuity for tasks. Although the primary goal of the project is to develop lighting countermeasures for the space environment, the funded research project also has important implications for Earth-based applications including light therapy for treatment of circadian rhythm disorders (Advanced and Delayed Sleep Phase Syndromes, Shift-Work Disorder, and Jet-lag Disorder) and circadian rhythm and sleep disorders caused by unusual light-dark environments such as Antarctica and in submarines. Light is also being developed as a non-pharmaceutical fatigue countermeasure in many other settings where high alertness and performance is paramount,

	such as military personnel, first-responders (police, firefighters, paramedics), and other safety-sensitive professions (e.g., pilots, physicians, nurses, truck drivers). Blue- or blue-enriched light therapy is also being tested for a number of affective and other clinical disorders to treat depression in Seasonal Affective Disorder, daytime fatigue in Traumatic Brain Injury, and to slow the rate of cognitive decline and depression in dementia. Our work has already begun to influence architectural lighting design with the development of methods both by ourselves and others to incorporate the non-visual effects of light into design of healthy buildings, aimed at reinforcing robust circadian entrainment through light-dark cycles, and the potential of light to enhance workplace performance and alertness. Dr. Lockley is also a member of several national and international Committees tasked with developing new lighting standards for the non-visual effects of light which will be informed by the current project and will ultimately be used by industries world-wide to develop lighting and lighting practices that optimize both the visual and non-visual effects of light.
Task Progress:	Space missions often expose crew members to unusual light-dark cycles that lead to misalignment between the circadian pacemaker and the sleep-wake schedule, resulting in disturbed sleep and impaired waking function. Light exposure is an effective countermeasure to facilitate more rapid adaptation of the circadian system and directly enhance alertness and performance. Recently, we and others have shown that short-wavelength (blue) light exposure at night is the most effective wavelength for phase-shifting the circadian pacemaker, suppressing pineal melatonin, enhancing subjective alertness, improving performance, and inducing electroencephalogram (EEG)-derived brain activation that indicates a more alert state. Before blue light is operationalized, the proposed project will test whether the spectral sensitivity of the circadian photoreception system is intensity-dependent, response-dependent, or time-of-day dependent. We aim to test the hypotheses that: 1) exposure to 6.5 of moderately dim monochromatic green (555 nm) light during the biological night will cause a greater circadian phase shifting, melatonin suppression, and arousal response than exposure to an equal photon density (2.8 x 1011 photons/cm2/s) of monochromatic blue (460 nm) light. 2) exposure to 6.5 of moderately bright monochromatic 460 nm light during the biological day will cause a greater circadian phase shifting and arousal response (but not enhancement of EEG-derived high-alpha activity [10.5-12 Hz]), than exposure to an equal photon density (2.8 x 1013 photons/cm2/s) of monochromatic 555nm light. 3) exposure to intermittent monochromatic 555 nm light (5 cycles x 2 mins light every 78 mins) for 6.5 hours during the biological night will cause a greater circadian phase shifting response than exposure to an equal photon density (2.8 x 1013 photons/cm2/s) of continuous monochromatic 555 nm light. We have completed Experiment 1 and 18 subjects completed the protocol. We had some technical problems with the monochromatic light system required to generate the controlled lighting for the study which delayed recruitment temporarily. We took extra steps to reduce future delays by purchasing a new monochromatic light generation device using other federal funds. We have completed Experiment 2 and 17 subjects completed the protocol. We have completed Specific Aim 3 and 6 subjects completed the protocol. Nine additional subjects were admitted to the inpatient study but withdrew or were disempanelled during the study. Data analysis is complete for the results for phase shifting, melatonin suppression, subjective alertness, and psychomotor performance from Experiments 1 and 2 (see Main Findings), and partially complete for phase shifting and performance in Experiment 3. We are currently analyzing waking EEG data from Experiments 1-3 and anticipate completion of EEG analysis shortly.
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